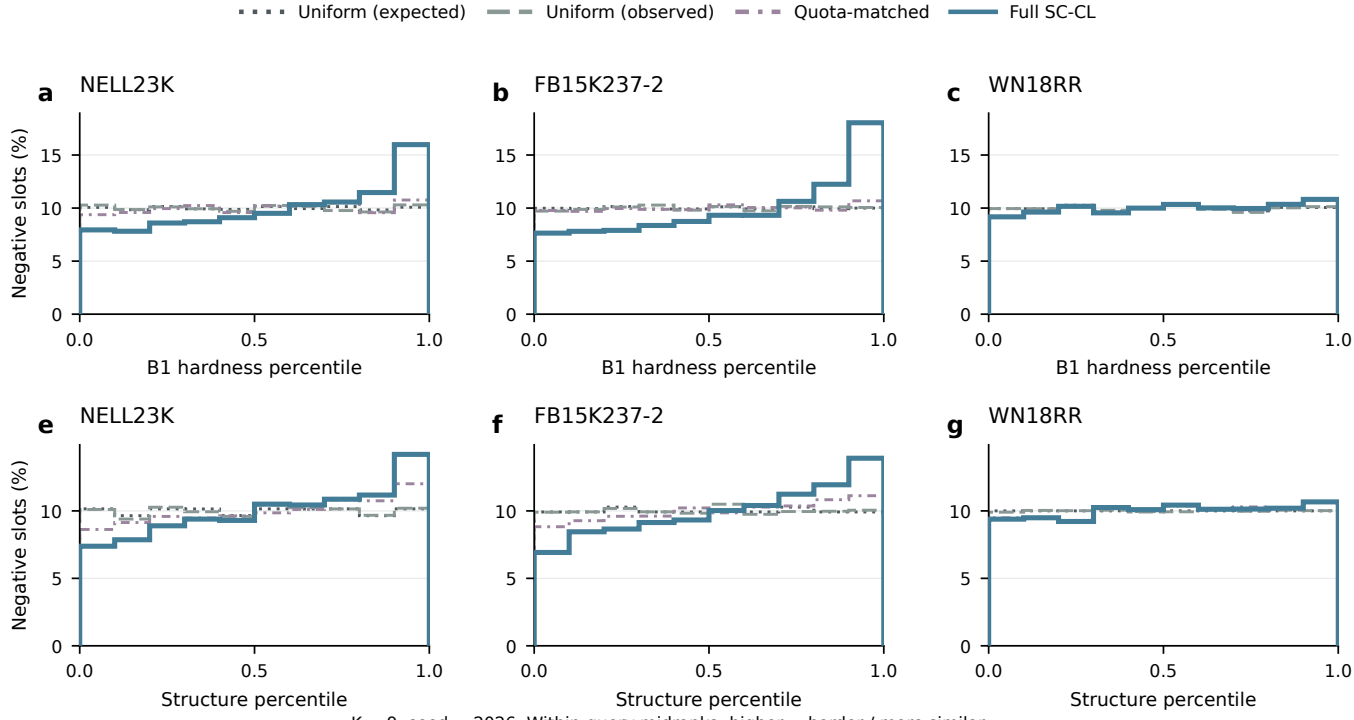
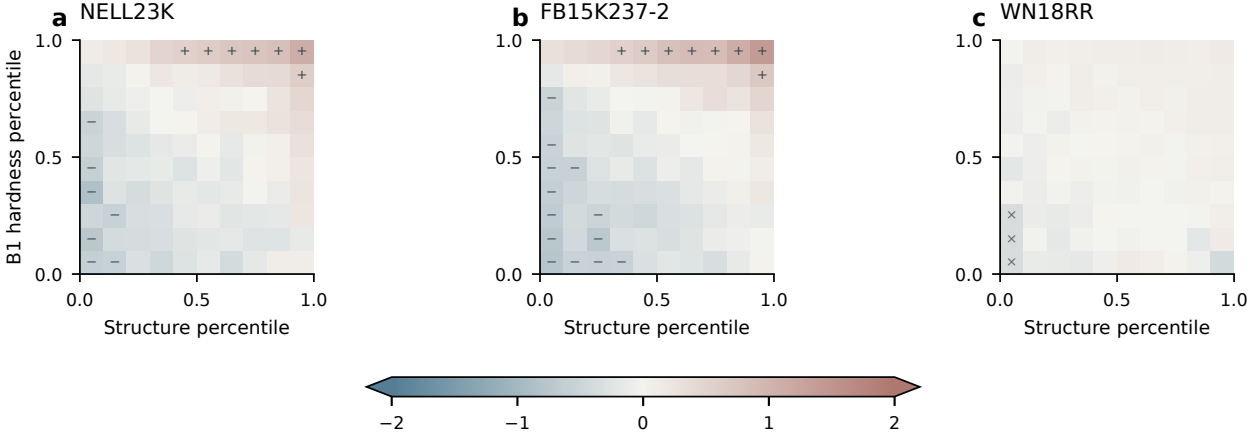
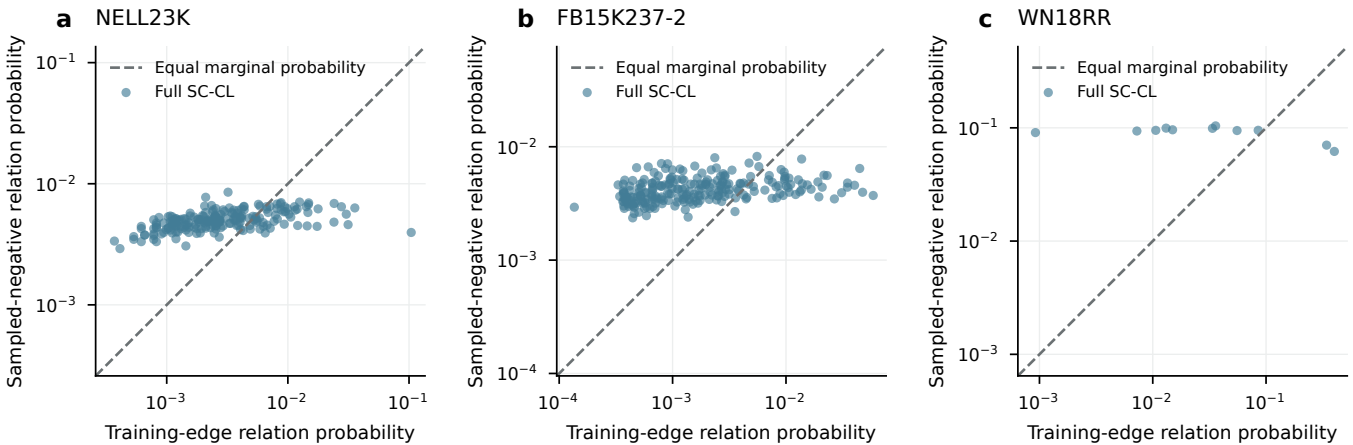


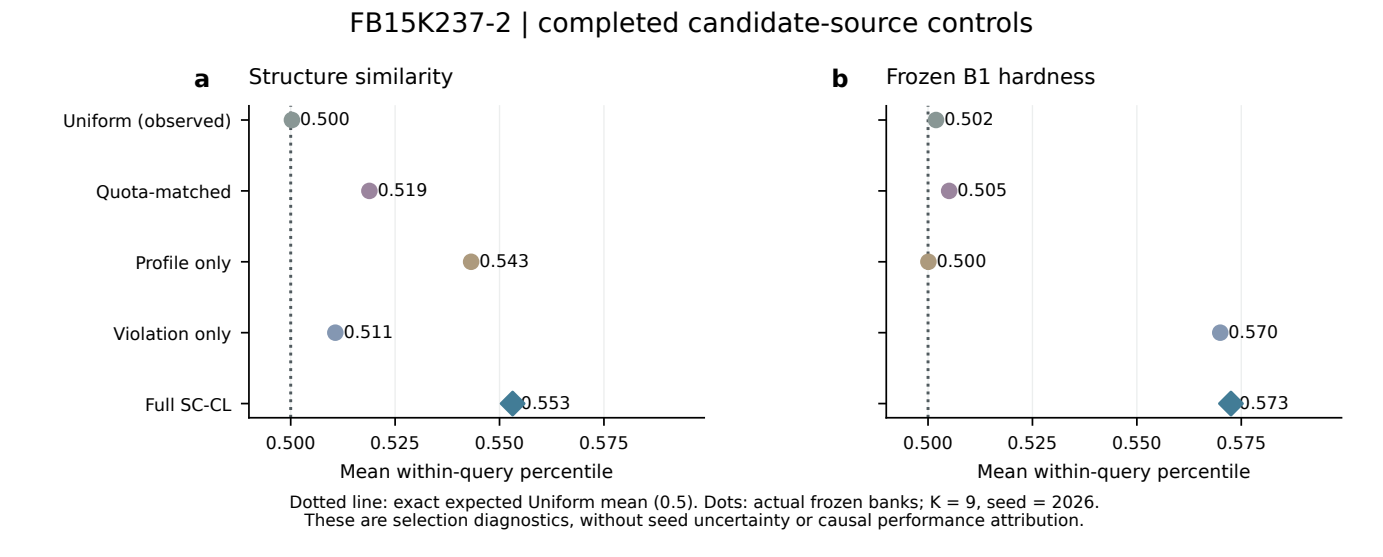
K = 9; seed = 2026. Within-query midranks; higher = harder / more similar.
Three datasets with aligned observed controls; WN18RR retains 55 unverified historical pairs.

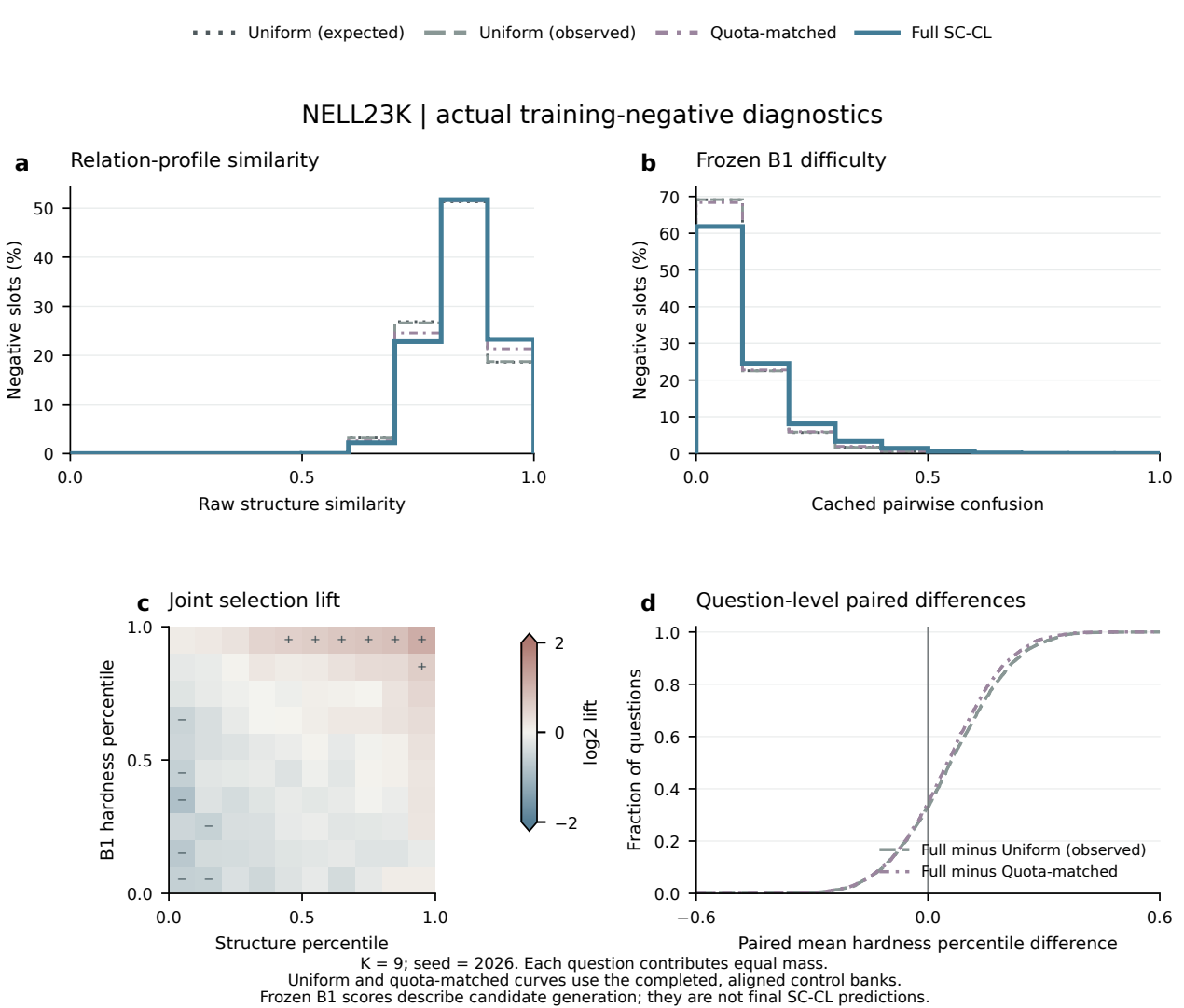


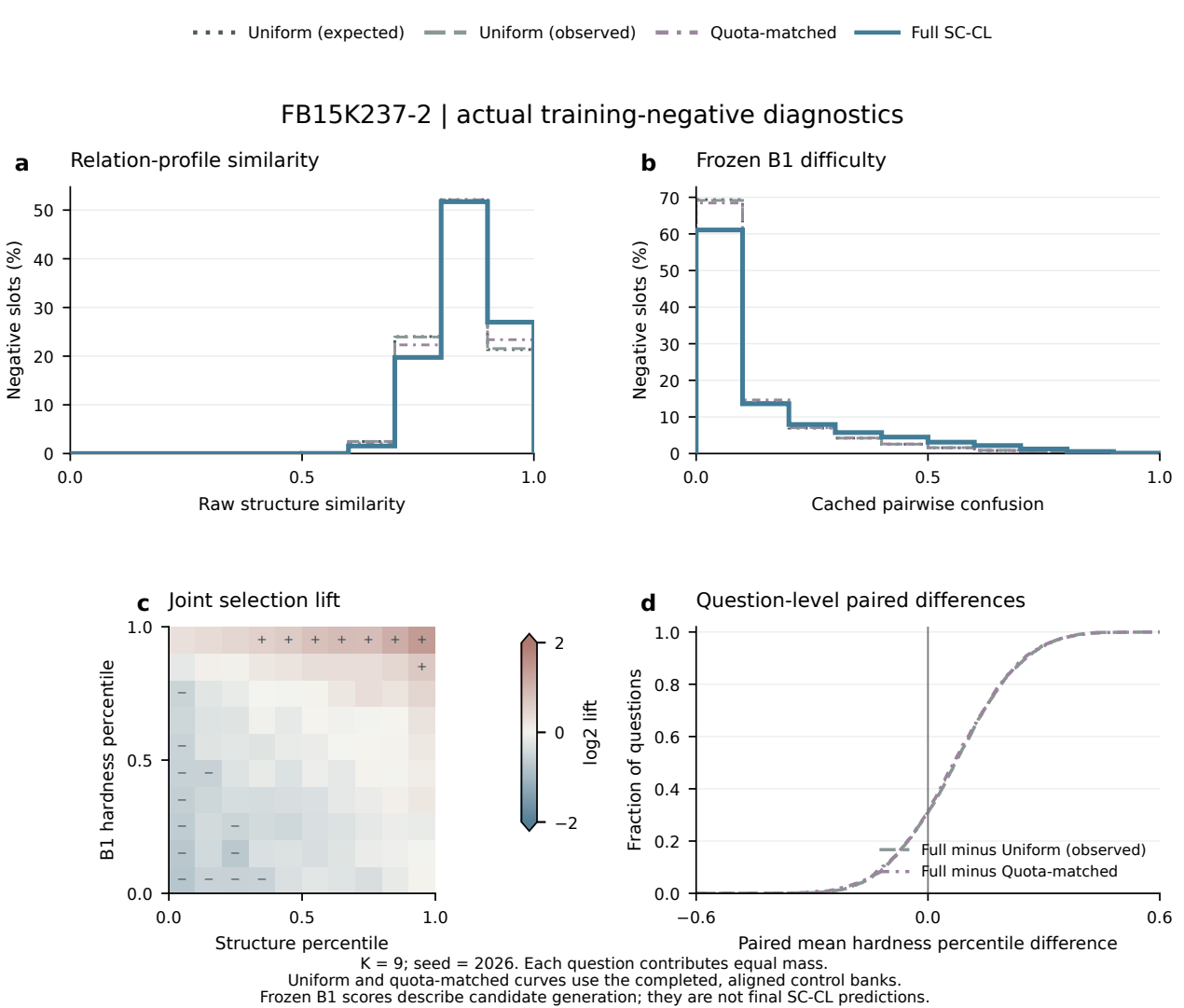
Gray ×: no eligible mass. Signs: |log2 lift| ≥ 0.5. Zero Full mass uses the lower extension.
WN18RR retains historical pools; selection enrichment is descriptive.

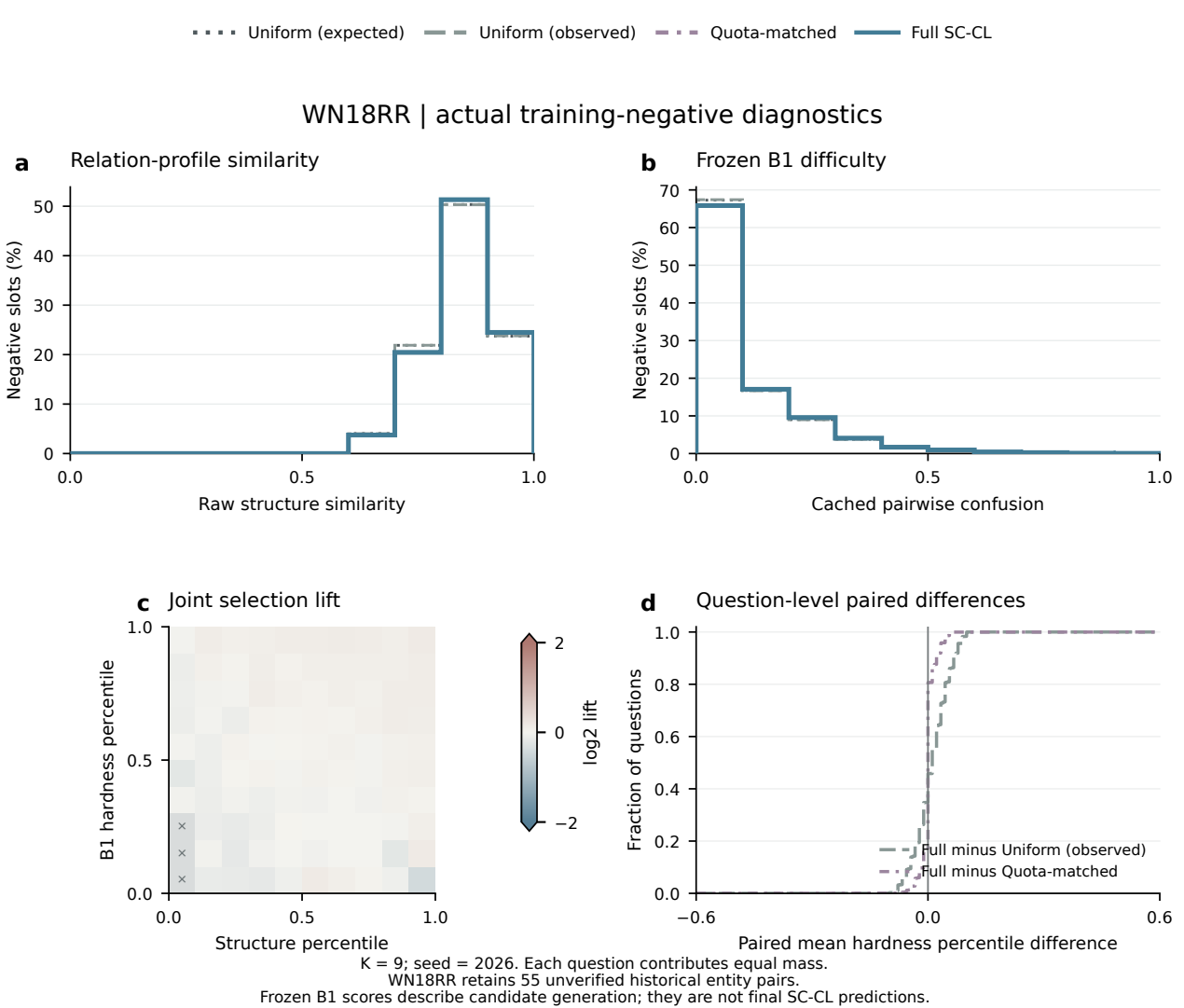


Each dot is one mapped relation label. Both axes use log scales.
The equality line is descriptive; no power-law fit or original-distribution matching objective is assumed.


FB15K237-2 | completed candidate-source controls
a
Structure similarity
Uniform (observed)
Quota-matched
Profile only
Violation only
Full SC-CL
0.500
0.519
0.543
0.511
0.553
0.500
0.525
0.550
0.575
Mean within-query percentile
b
Frozen B1 hardness
0.502
0.505
0.500
0.570
0.573
0.500
0.525
0.550
0.575
Mean within-query percentile
Dotted line: exact expected Uniform mean (0.5). Dots: actual frozen banks; K = 9, seed = 2026.
These are selection diagnostics, without seed uncertainty or causal performance attribution.


Uniform (expected)
Uniform (observed)
Quota-matched
Full SC-CL
NELL23K | actual training-negative diagnostics
a
Relation-profile similarity
Negative slots (%)
Raw structure similarity
b
Frozen B1 difficulty
Negative slots (%)
Cached pairwise confusion
c
Joint selection lift
B1 hardness percentile
Structure percentile
log2 lift
d
Question-level paired differences
Fraction of questions
Full minus Uniform (observed)
Full minus Quota-matched
Paired mean hardness percentile difference
K = 9; seed = 2026. Each question contributes equal mass.
Uniform and quota-matched curves use the completed, aligned control banks.
Frozen B1 scores describe candidate generation; they are not final SC-CL predictions.


Uniform (expected)
Uniform (observed)
Quota-matched
Full SC-CL
FB15K237-2 | actual training-negative diagnostics
a
Relation-profile similarity
Negative slots (%)
Raw structure similarity
b
Frozen B1 difficulty
Negative slots (%)
Cached pairwise confusion
c
Joint selection lift
B1 hardness percentile
Structure percentile
log2 lift
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Question-level paired differences
Fraction of questions
Full minus Uniform (observed)
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Paired mean hardness percentile difference
K = 9; seed = 2026. Each question contributes equal mass.
Uniform and quota-matched curves use the completed, aligned control banks.
Frozen B1 scores describe candidate generation; they are not final SC-CL predictions.


Uniform (expected)
Uniform (observed)
Quota-matched
Full SC-CL
WN18RR | actual training-negative diagnostics
a
Relation-profile similarity
Negative slots (%)
Raw structure similarity
b
Frozen B1 difficulty
Negative slots (%)
Cached pairwise confusion
c
Joint selection lift
B1 hardness percentile
Structure percentile
log2 lift
d
Question-level paired differences
Fraction of questions
Full minus Uniform (observed)
Full minus Quota-matched
Paired mean hardness percentile difference
K = 9; seed = 2026. Each question contributes equal mass.
WN18RR retains 55 unverified historical entity pairs.
Frozen B1 scores describe candidate generation; they are not final SC-CL predictions.